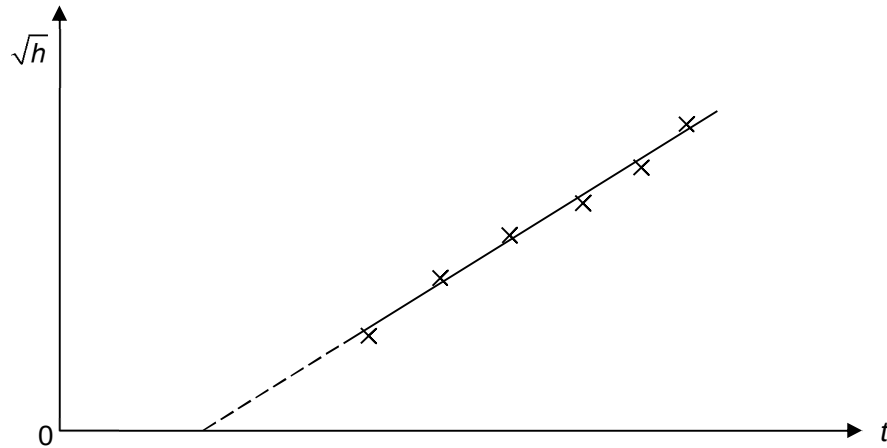


9	B	19	D	29	D
10	C	20	B	30	A

- 1 A student measures the time t for a ball to fall from rest through a vertical distance h . The student plots his results and best-fit line in the graph shown.



Which of the following statement is true?

- A The result is accurate as the line is close to the data points
- B The result is not accurate as the line does not pass through the origin
- C Data is precise as there are equal number of data points on both sides of the line
- D Data is precise as the data points do not deviate from the line

Ans: (B)

At time = 0, the height fallen should be zero since the ball is still at the starting point.

- 2 The experimental measurement of the heat capacity of a solid as a function of temperature T is found to fit the following expression

$$C = \alpha T^3 + \beta T$$

What are the possible base units of α and β ?

units of α

units of β

- | | | |
|----------|---|---|
| A | $\text{kg m}^2 \text{s}^{-1} \text{K}^{-4}$ | $\text{kg m}^2 \text{s}^{-1} \text{K}^{-1}$ |
| B | $\text{kg}^2 \text{m s}^{-2} \text{K}^{-3}$ | $\text{kg}^2 \text{m s}^{-2} \text{K}^{-2}$ |
| C | $\text{kg m}^2 \text{s}^{-2} \text{K}^{-4}$ | $\text{kg m}^2 \text{s}^{-2} \text{K}^{-2}$ |
| D | $\text{kg}^2 \text{m s}^{-2} \text{K}^{-3}$ | $\text{kg}^2 \text{m s}^{-2} \text{K}^{-1}$ |

Ans: (C)

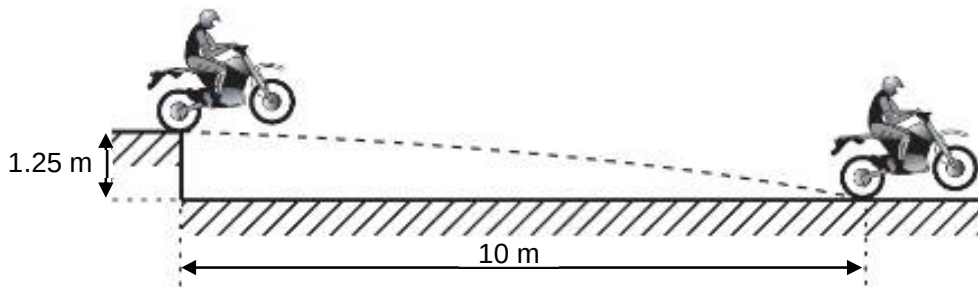
$$C = \alpha T^3 + \beta T$$

$$\text{J K}^{-1} = [\alpha](\text{K}^3) = [\beta](\text{K})$$

$$\text{Using } E = \frac{1}{2}mv^2 \quad \text{J} = \text{kg m}^2 \text{s}^{-2}$$

$$\text{kg m}^2 \text{s}^{-2} \text{K}^{-1} = [\alpha](\text{K}^3) = [\beta](\text{K})$$

- 3** A motorcycle stunt-rider moving horizontally takes off from a point 1.25 m above the ground, landing 10 m away as shown in the diagram.



What was the speed at take-off?

- A** 5 m s⁻¹ **B** 10 m s⁻¹ **C** 15 m s⁻¹ **D** 20 m s⁻¹

Ans: D

$$\downarrow s_y = u_y t + \frac{1}{2} a_y t^2 \Rightarrow 1.25 = 0 + \frac{1}{2} (9.81) t^2 \Rightarrow t = 0.50 \text{ s}$$

$$\rightarrow s_x = u_x t \Rightarrow 10 = u_x (0.50) \Rightarrow u_x = 20 \text{ m s}^{-1}$$

- 4** A body of mass 3.0 kg is thrown with a velocity of 20 m s⁻¹ at an angle of 60° above horizontal. It reaches the maximum height after 1.8 s. Air resistance is negligible.

What is the rate of change of momentum of the body at the maximum height?

- A** zero **B** 17 kg m s⁻² **C** 29 kg m s⁻² **D** 33 kg m s⁻²

Ans: C

$$\text{rate of change of momentum} = \text{net force} = \text{weight} = 3.0 \times 9.81 = 29 \text{ N}$$

$$\text{OR use } (mv_y - mu_y)/t = (0 - 3.0 \times 20 \sin 60^\circ) / 1.8 = -29 \text{ kg m s}^{-2}$$

- 5** A body P of mass 2.0 kg and moving with velocity +3.0 m s⁻¹ makes a head-on inelastic collision with a stationary body Q of mass 4.0 kg.

Which of the following could be the velocities of P and Q after the collision?

[Turn over

	velocity of P after collision	velocity of Q after collision
A	$+0.5 \text{ m s}^{-1}$	$+0.5 \text{ m s}^{-1}$
B	$+0.0 \text{ m s}^{-1}$	$+3.0 \text{ m s}^{-1}$
C	-1.0 m s^{-1}	$+2.0 \text{ m s}^{-1}$
D	-0.6 m s^{-1}	$+1.8 \text{ m s}^{-1}$

Ans: D

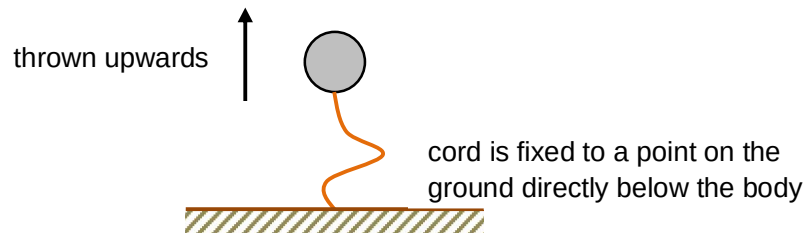
Initial momentum = $2.0 \times 3.0 = +6.0 \text{ Ns}$

Final momentum = $2.0(-0.6) + 4.0(1.8) = -1.2 + 7.2 = +6.0 \text{ Ns}$

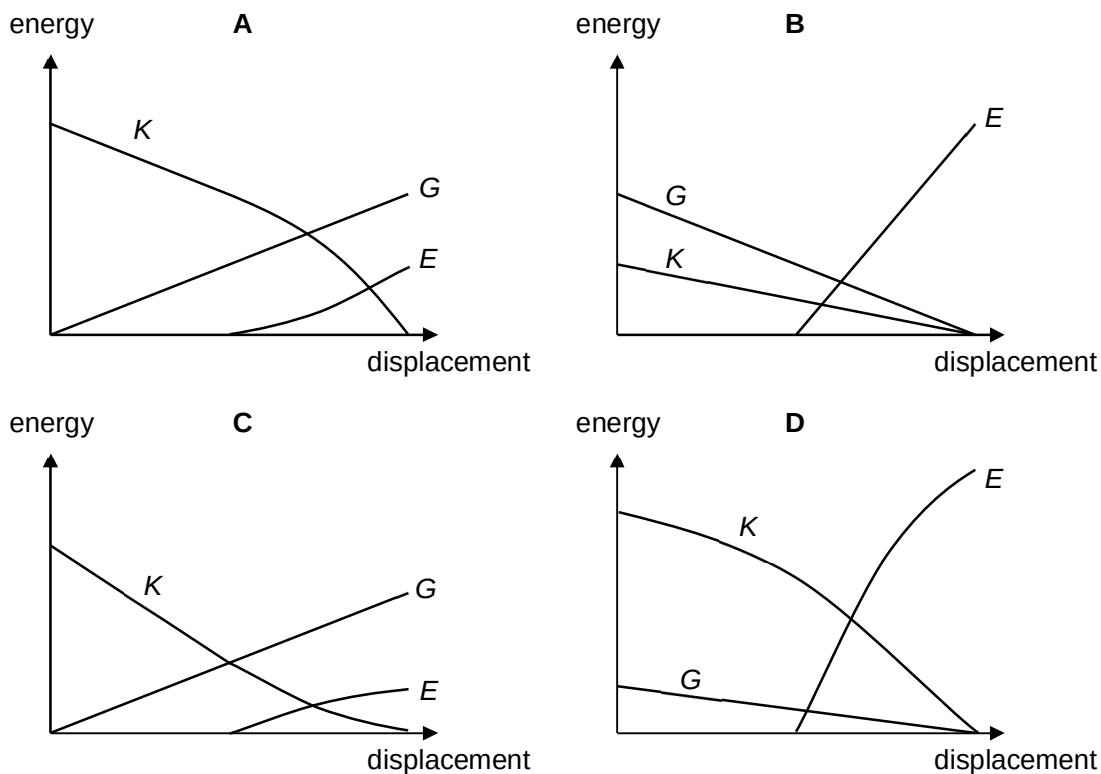
only D has momentum conserved but KE not conserved (or has lesser relative speed of separation than relative speed of approach)

Both momentum and KE are conserved for C (elastic collision)

- 6 The diagram shows a body attached to an elastic cord being thrown vertically upwards. Initially the cord is unstretched but after a while it becomes stretched. The cord obeys Hooke's law and air resistance is ignored.



Which of the following shows the variation with displacement of the kinetic energy K , gravitational potential energy G and elastic potential energy E ?



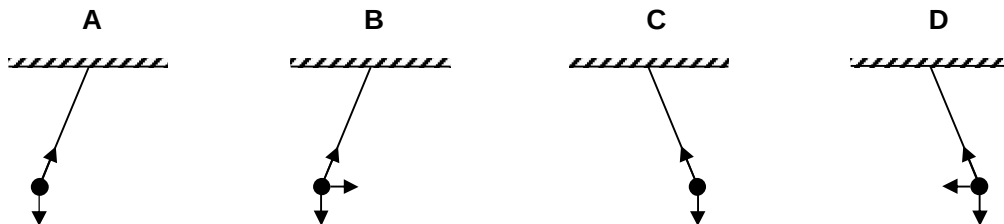
Ans: A

GPE increases linearly with displacement

EPE starts later, increases with extension squared (parabolic upwards)

KE decreases linearly with displacement at first (vs GPE), then decreases at greater and greater rate due to increasing EPE

- 7 A passenger is sitting in a railway carriage facing in the direction in which the train is travelling. A pendulum hangs down in front of him from the carriage roof. The train travels along a circular arc bending to the left. Which one of the following diagrams shows the position of the pendulum as seen by the passenger, and the directions of the forces acting on it?



Ans: C

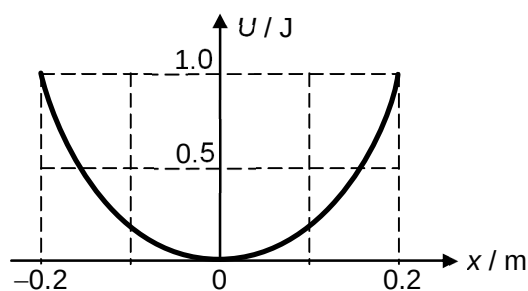
- 8 In two widely-separated planetary systems whose suns have masses S_1 and S_2 , planet P_1 of mass M_1 (orbiting sun S_1) and planet P_2 of mass M_2 (orbiting sun S_2) are observed to have circular orbits of equal radii. If P_1 completes an orbit in half the time taken by P_2 , it may be deduced that

[Turn over

- A $S_1 = S_2$ and $M_1 = 0.25 M_2$
- B $S_1 = 4S_2$ only
- C $S_1 = 4S_2$ and $M_1 = M_2$
- D $S_1 = 0.25 S_2$ only

Ans: (B)

- 9 A particle of mass 4.0 kg moves in simple harmonic motion. Its potential energy U varies with position x as shown in the figure below.



What is the period of oscillation of the mass?

- A $\frac{2\pi}{25}$ s
- B $\frac{2\pi\sqrt{2}}{5}$ s
- C $\frac{8\pi}{25}$ s
- D $\frac{4\pi}{5}$ s

Ans B

Total energy of system is constant.

$$\text{Max PE} = 1.0 \text{ J} = \text{Max KE} = \frac{1}{2} m v_{\text{max}}^2$$

$$v_{\text{max}} = \frac{1}{\sqrt{2}}$$

$$v_{\text{max}} = \omega X_o = \frac{2\pi}{T} X_o$$

$$\begin{aligned} T &= \left(\frac{2\pi}{v_{\text{max}}} \right) X_o \\ &= 2\pi (\sqrt{2})(0.2) \\ &= \frac{2\pi\sqrt{2}}{5} \end{aligned}$$

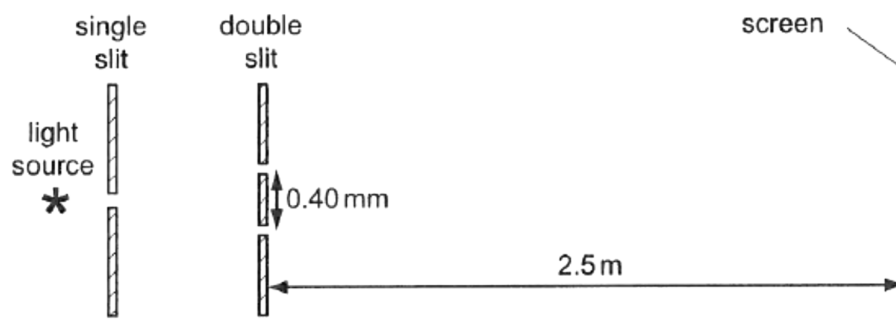
- 10 A toy car moving along a horizontal plane in simple harmonic motion starts from the amplitude at time $t = 0$ s. If the amplitude of its motion is 5.0 cm and frequency is 2.0 Hz, the magnitude of the acceleration of the toy car at 1.7 s is

- A 0.25 m s^{-2}
- B 0.51 m s^{-2}
- C 6.4 m s^{-2}
- D 7.4 m s^{-2}

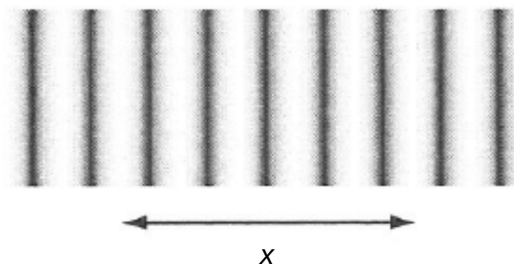
Ans C

$$\begin{aligned}
 x &= x_o \cos(\omega t) \\
 &= 5.0 \cos(2\pi(2) \times 1.7) \\
 &= -4.05 \text{ cm} \\
 a &= |\omega^2 x| \\
 &= (4\pi)^2 (4.05 \times 10^{-2}) \\
 &= 6.4 \text{ m s}^{-2}
 \end{aligned}$$

- 11 A two source interference experiment is set up as shown.



The source emits light of wavelength 600 nm. The interference pattern on the screen is shown below.



What is the distance x ?

- A $3.8 \times 10^{-4} \text{ m}$ B $1.9 \times 10^{-3} \text{ m}$ C $3.8 \times 10^{-3} \text{ m}$ D $1.9 \times 10^{-2} \text{ m}$

Ans D

$$\Delta x = \frac{\lambda D}{a} = \frac{600 \times 10^{-9} (2.5)}{0.40 \times 10^{-3}} = 3.75 \times 10^{-3} \text{ m}$$

$$x = 5\Delta x = 5(3.75 \times 10^{-3}) = 1.9 \times 10^{-2} \text{ m}$$

- 12 A guitar string of length L is stretched between two fixed points P and Q and made to vibrate